

Introduction to Software Testing From Manual to AI-Powered QA



Rashmi Ruwanthika Athukorala · 3 min read · Just now



Software testing plays a very important role in building quality software. It helps us find bugs, check if everything works as expected, and make sure the final product gives a good user experience.

In the early days, testing was done **manually**. Testers used to check every function, page, and feature by themselves. Manual testing helps understand how real users interact with the system, but it also takes a lot of time and can be tiring for large projects.

To make testing faster and more reliable, **automation testing** was introduced. Automation allows us to write test scripts that can run automatically and repeatedly without human effort. This is where tools like **Selenium** and frameworks like **JUnit** became very useful.

Selenium — The Web Automation Tool

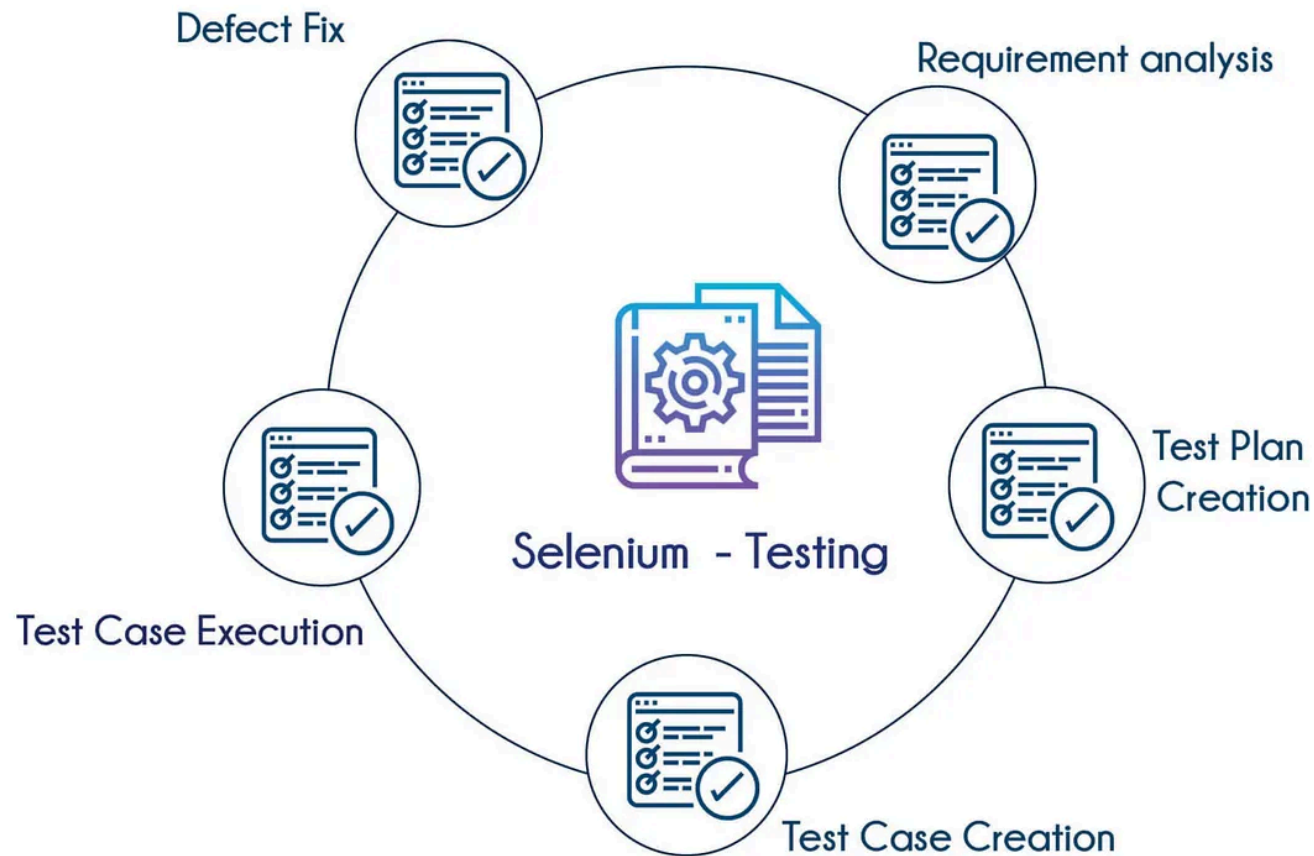
Selenium is one of the most popular tools for automating web applications. It allows testers to write scripts in many programming languages, such as Java, to perform actions like clicking buttons, filling forms, and checking if web pages load correctly.

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By using Selenium, we can test websites more quickly and cover more test cases compared to manual testing. It helps find bugs early and ensures that web applications work the same way across different browsers.



JUnit — The Testing Framework

JUnit is a testing framework mainly used with Java. It helps testers create small, reusable test cases to check if each part of the program works correctly.

When used together with Selenium, JUnit helps organize and manage test scripts better. It also provides detailed reports that show which tests passed

or failed, making it easier to track results and fix issues. Moving Towards AI Powered QA

Today, the world of software testing is changing again. With the help of **AI (Artificial Intelligence)**, testing is becoming smarter. AI can analyze results, predict failures, and even generate test cases automatically.

But before we reach that level, learning the basics of **Selenium and JUnit** is very important. They are the foundation that helps QA engineers understand the real process of automation.

Software testing has come a long way from manual checking to smart automation and now towards AI powered quality assurance.

Tools like **Selenium** and **JUnit** are still the heart of modern testing and a great starting point for anyone who wants to build a strong career in QA. Software testing is one of the most important steps in software development. It helps to make sure that the application works correctly, safely, and smoothly before it reaches the users. As technology grows, the way we test software has also evolved from **manual testing** to **automated** and now **AI assisted testing**.

Software Testing



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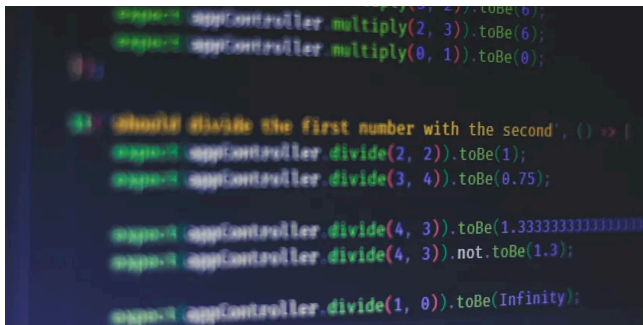


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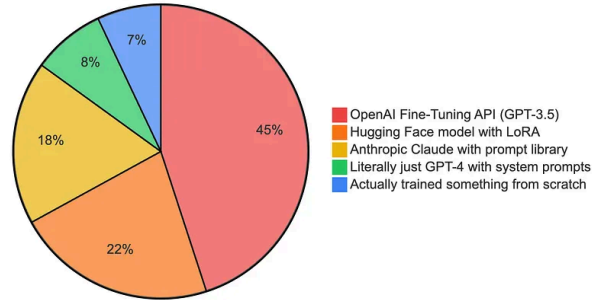
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